07.12.2025

Logistic Regression

CSE 329

Linear Regression

- Given x, we approximate y with a linear function.

$$h(x) = \sum_{i=0}^{d} \theta_i x_i = \theta^T x$$

- θ is called parameters/weights

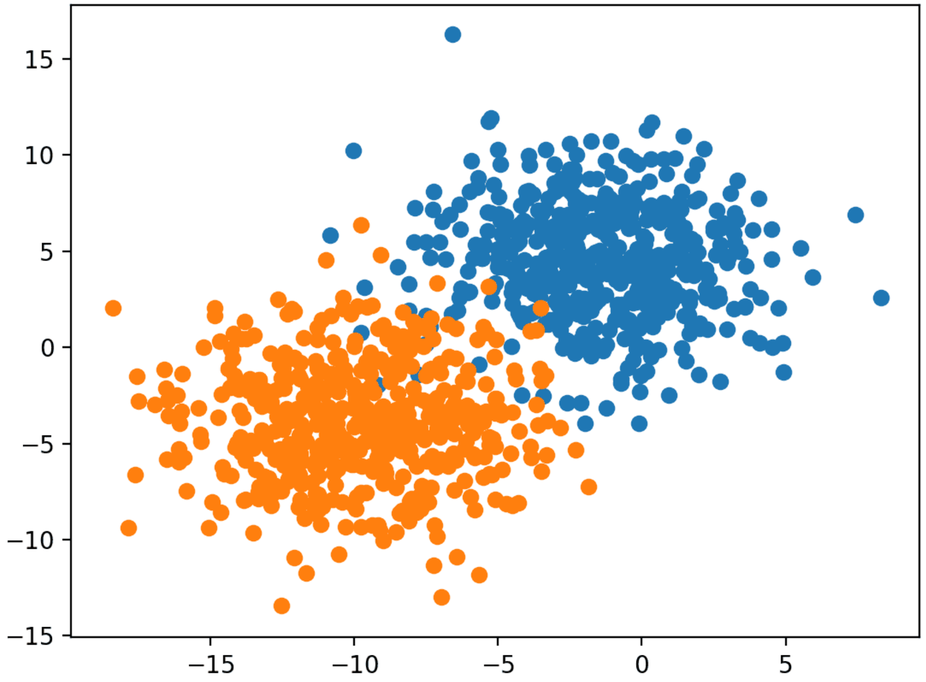
- What about classification?

Classification

- Y is a discrete target space
- Commonly binary classification tasks
- $y \in \{0, 1\}$
- Often called positive class vs negative class

1 → 1

−1 → 0



যেকোনো একটা value 1 হতে হবে

Logistic Regression

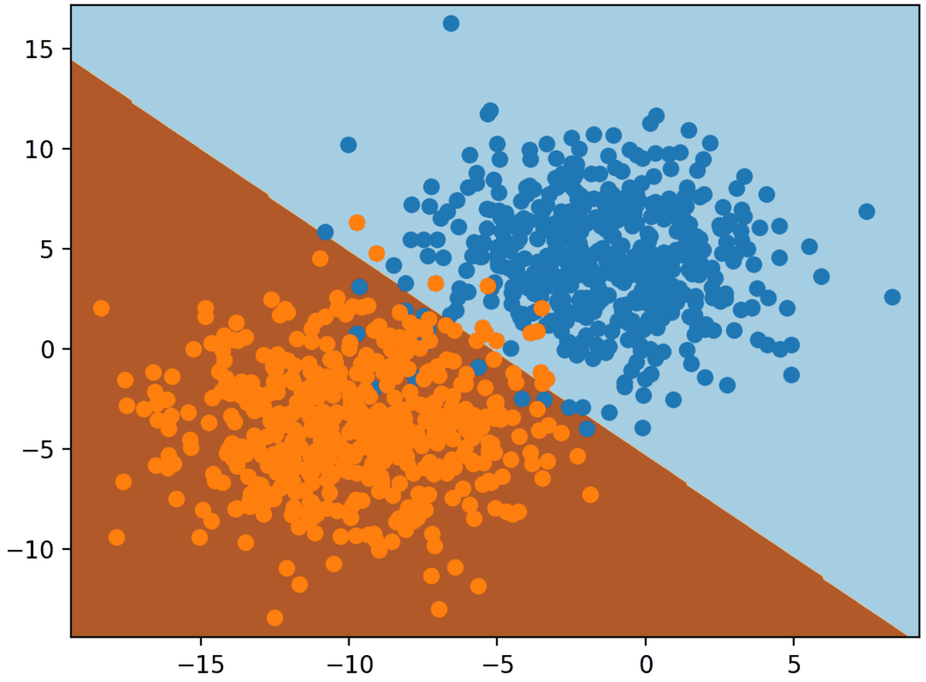
- Instead of predicting $y \in \{0, 1\}$, we can predict $\mathbb{P}(y = 1)$

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Why this function?

$P(y=1) > 0.5 \rightarrow$ sample 1 থেকে আসছে predict করবে



Classification with Linear Classifier

	To Predict	Range		
		Min	Middle	Max
Binary Classification	y	0	-	1
Predict Probability	$p = P(y=1), q = 1-p$	0	0.5	1
Odds	$\frac{p}{1-p}$	0	1	∞
Log-odds	$\log\left(\frac{p}{1-p}\right)$	$-\infty$	0	∞
Linear Classifier Prediction		$-\infty$	0	∞

continuous value

confused (কোন class এ পড়ছে বলতে পারবো না)

↳ যেকোনো range এ করতে পারি $(-\infty, \infty)$

Logistic Regression

(x, y) data দেখায়

$y = f(x, \theta)$

→ কোন θ হলে এই data হওয়ার probability high সেটা বের করবো

$$h_\theta(x) = \theta^T x = \log\left(\frac{p}{1-p}\right)$$

$$e^{\theta^T x} = \frac{p}{1-p}$$

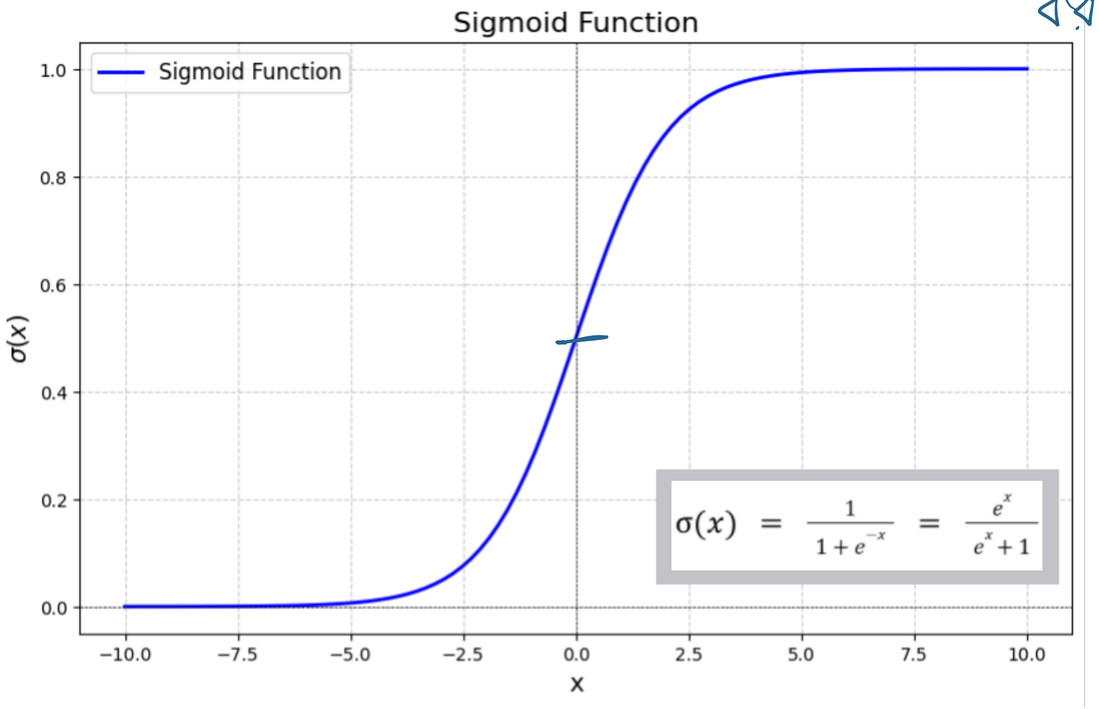
$$e^{\theta^T x} - e^{\theta^T x} p = p$$

$$e^{\theta^T x} = p(1 + e^{\theta^T x})$$

$$p = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}}$$

$$p = \frac{1}{1 + e^{-\theta^T x}}$$

$$P(y = 1) = p = \sigma(\theta^T x)$$



$x = 0$ $\quad 0.5$

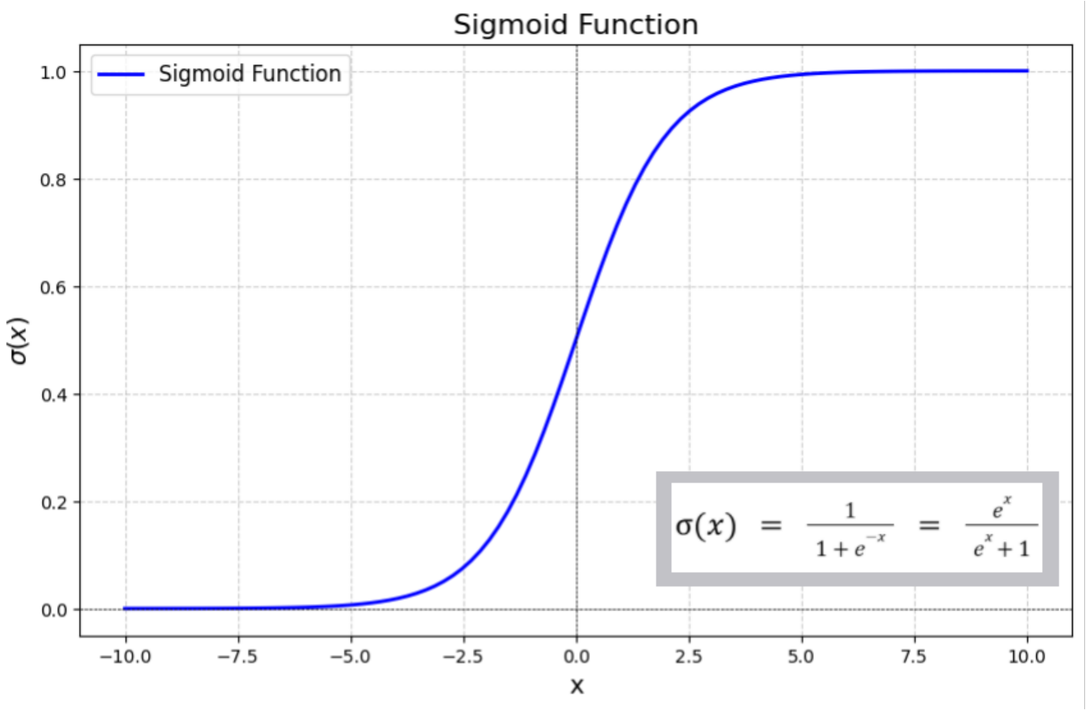
$x > 0$ $\quad > 0.5$

$x < 0$ $\quad < 0.5$

Sigmoid Function

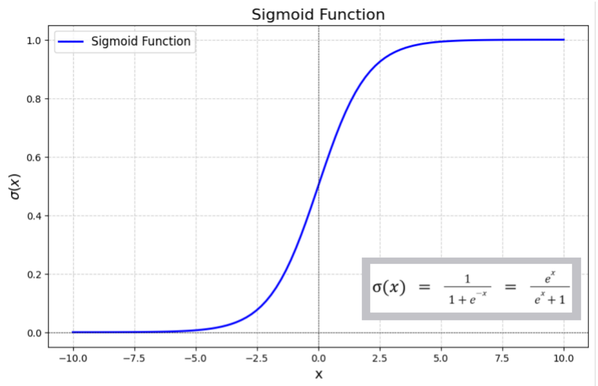
- Some properties,

$$g'(z) = g(z)(1 - g(z)).$$



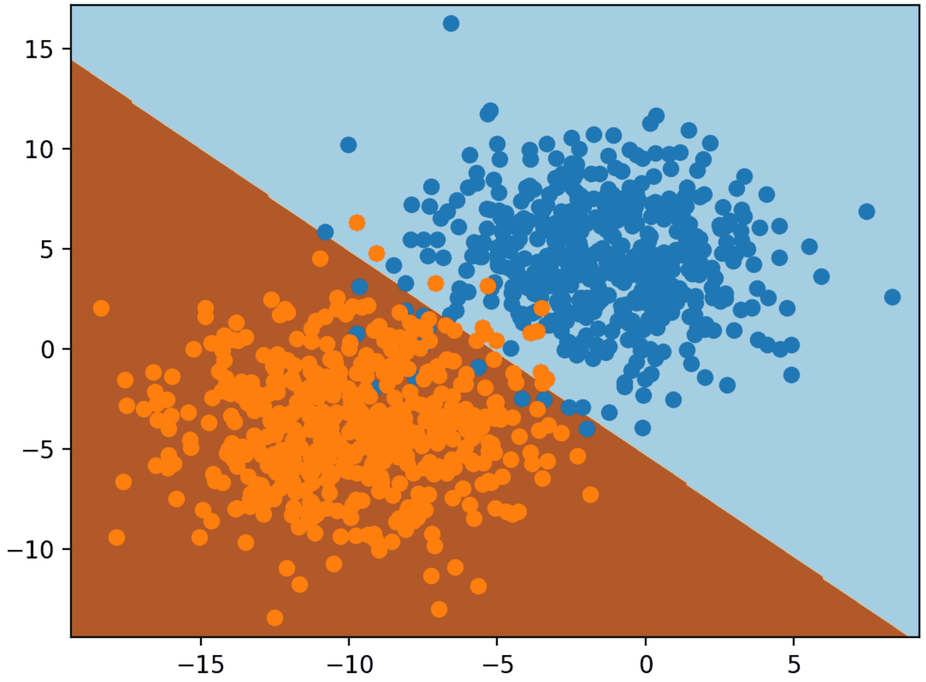
Logistic Regression

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$



- How to choose θ?

- Maximum Likelihood Estimation



Probability (Recap)

Bayes' Theorem

- Given two events A and B,

$$\mathrm{P}(A, B) = \mathrm{P}(A|B)\mathbb{P}(B) = \mathrm{P}(B|A)\mathrm{P}(A)$$

and,

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}$$

Probability (Recap)

Bayes' Theorem

- Given (training) data D and a parameter θ $(random\ variable)$,
- Likelihood, $\mathrm{P}(D|\theta)$
- Prior, $\mathrm{P}(\theta)$
- Evidence, $\mathrm{P}(D)$
- Posterior,

$$\mathrm{P}(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Posterior $\propto$ Likelihood $\times$ Prior

so likelihood maximize করবো।

D given, কোন θ হওয়া সবচেয়ে probable?

P(θ|D) → maximize করতে চাই

P(D) → fixed

P(θ) → fixed as hypo. fix করে ফেলছি (linear)

Assume, $\theta \rightarrow$ comes from some space

$\hookrightarrow$ hypothesis space (θ can be any value from this space)

$\theta \rightarrow$ uniformly distributed

যেকোনো θ হওয়া equally likely.

$P(\theta)$ <u>fixed</u>

$\hookrightarrow$ কোনোরকম modify করবো না

$$y = f(x; \theta)$$

$P(D|\theta)$ এমন θ চাই যার জন্য এই data (D) appear করার probability high.

$$P(x, y \mid \theta)$$
$$= P(y \mid x, \theta)\, \underline{P(x \mid \theta)}$$

$P(y|x, \theta)$ maximize করার try করবো।

$y \rightarrow$ ground truth

$\theta \rightarrow$ only changes the relationship b/w x and y

Logistic Regression

- We assume

$$
\begin{aligned}
P(y=1 \mid x;\theta) &= h_\theta(x) \\
P(y=0 \mid x;\theta) &= 1 - h_\theta(x)
\end{aligned}
$$

Or,

$$
p(y \mid x;\theta) = (h_\theta(x))^y \, (1 - h_\theta(x))^{1-y}
$$

- Likelihood

$$
\begin{aligned}
L(\theta) &= p(\vec{y} \mid X;\theta) \\
&= \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)};\theta) \\
&= \prod_{i=1}^{n} \left(h_\theta(x^{(i)})\right)^{y^{(i)}} \left(1 - h_\theta(x^{(i)})\right)^{1-y^{(i)}}
\end{aligned}
$$

sample গুলো iid ছিল
তাই multiply করা
লাগছে

Logistic Regression

- Likelihood

$$
\begin{aligned}
L(\theta) &= p(\vec{y} \mid X; \theta) \\
&= \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)}; \theta) \\
&= \prod_{i=1}^{n} \left(h_\theta(x^{(i)})\right)^{y^{(i)}} \left(1 - h_\theta(x^{(i)})\right)^{1-y^{(i)}}
\end{aligned}
$$

- Log Likelihood

$$
\begin{aligned}
\ell(\theta) &= \log L(\theta) \\
&= \sum_{i=1}^{n} y^{(i)} \log h(x^{(i)}) + (1 - y^{(i)}) \log(1 - h(x^{(i)}))
\end{aligned}
$$

Logistic Regression

- Maximize log likelihood with,
  - Stochastic Gradient Ascent

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

→ কত তম sample

→ কত তম feature

  - Gradient Ascent
    (Batch)

$$\theta_j := \theta_j + \frac{\alpha}{m}? \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

sigmoid function apply করা necessary না।

Perceptron

↳ DLG use হয়

- We can modify the logistic regression method to “force” it to output values that are either 0 or 1 or exactly.

$$h_\theta(x) = g(\theta^T x)$$

- Where

$\theta^T x$

$$g(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

- With the same update rule,

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

Reference

- CS229 Lecture Note
  - Section 2.1